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Sub queries

- Sub-query is query within query. Typically it work with SELECT statements.
- Output of inner query is used as input to outer query.
- If no optimization is enabled, for each row of outer query result, sub-query is executed once. This reduce performance of sub-query.
- Single row sub-query
 - Sub-query returns single row.
 - Usually it is compared in outer query using relational operators.



Sub queries

- Multi-row sub-query
 - Sub-query returns multiple rows.
- Usually it is compared in outer query using operators like IN, ANY or ALL.
 - IN operator checks for equality with results from sub-queries.
 - ANY operator compares with one of the result from sub-queries.
 - ALL operator compares with all the results from sub-queries.



Sub queries

- Correlated sub-query
 - If number of results from sub-query are reduced, query performance will increase.
 - This can be done by adding criteria (WHERE clause) in sub-query based on outer query row.
 - Typically correlated sub-query use IN, ALL, ANY and EXISTS operators.



Sub query

- Sub queries with UPDATE and DELETE are not supported in all RDBMS.
- In MySQL, Sub-queries in UPDATE/DELETE is allowed, but sub-query should not SELECT from the same table, on which UPDATE/DELETE operation is in progress.



Thank You

